Chapter 2

Locating the source of defective past tense marking in advanced L2 English speakers*

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1. Introduction

It is well-known that advanced L2 speakers of English from certain L1 backgrounds show persistent optionality in marking thematic verbs for simple past tense in spontaneous oral production, as for example in *The police caught the man and **take** him away*. Speakers whose L1 is Chinese appear to be such a group. Bayley (1991, 1996) found the phenomenon sufficiently robust in Chinese speakers to undertake a variationist analysis of the factors which might be causing it. Wolfram and Hatfield (1984) had found similar optionality in the L2 English of Vietnamese speakers. More recently Lardiere (1998a, 1998b, 2000) has reported remarkably consistent optionality in simple past tense marking in a near-native speaker of English sampled with an eight-and-a-half year interval. Patty, a native speaker of Mandarin and Hokkien, marked simple past tense on thematic verbs around only one-third of the time in data collected from spontaneous speech. Native speakers, by contrast, do not apparently show endemic optionality of the same kind, although failure to inflect a verb for past tense is found sporadically in 'slips of the tongue' (Fromkin 1988).[1]

An important question for theories of SLA which assume that the mental grammars of individual L2 speakers are derived from Universal Grammar (UG) is why such optionality might exist in advanced/near-native speakers. Given that the morphophonology of forms in English provides clear positive evidence that past tense is marked, it is unexpected that advanced/near-native speakers should continue to have problems with it. Locating the source of the difficulty would be a small contribution to the broader goal of determining exactly how the language faculty is involved in the construction of grammatical knowledge by older L2 learners.

Beck (1997) has argued that the kind of optionality in question is unlikely to be the result of a deficit in the component of the language faculty which generates inflected phonological word forms: morphology. Beck compared the reaction times of 31 non-native speakers from a variety of L1 backgrounds with those of 32 natives on a task requiring the production of past-inflected verb forms. Speakers were presented with verb stems on a computer screen, and required to produce the simple past tense form orally, which activated a timing device. In previous studies with natives (e.g. Prasada, Pinker and Snyder 1990) it had been found that irregular verb forms show a frequency effect: the more frequent the stem of the verb in question (where frequency is defined as frequency of occurrence in corpora of English usage) the faster the reaction time to the past tense form. For regular verbs, however, frequency of the stem form had no effect on reaction time. Such findings have led to the claim that irregular past tense verb forms are stored associatively in memory (i.e. are 'listed') and hence show frequency effects as a function of strength of association between the stem and the listed form. By contrast, regular inflection is produced by rule, which applies in the same way to all regular stems, independently of frequency. Hence there is no reaction time effect.

Beck's findings with the non-native speakers were that they performed similarly to the natives: reaction times on low frequency irregular stems were slower than on high frequency stems (although not significantly so), and there was no difference in their performance on frequent and less frequent regular stems. This led Beck to suggest that it is not the morphological component which is involved in causing persistent optionality in past tense marking.

Lardiere (1998a, 1998b, 2000) has suggested that the problem for Patty resides in the 'mapping' between fully specified syntactic phrase markers and surface morphophonology. Lardiere argues that other evidence from Patty's spontaneous oral production suggests that she has a T(ense) category specified for finiteness. Firstly, her use of nominative case-marked pronouns is perfect; on standard assumptions the nominative case of subjects in English is the result of an agreement or checking relation between a T category specified [+finite] and the subject in the specifier of TP. Hence if Patty uses nominative pronouns perfectly she must have represented a finite feature on T. Secondly, there is no evidence for thematic verb raising (e.g. over negation) in Patty's productions, suggesting that she has established that T has weak inflectional properties in English — another piece of evidence that Patty has a T specified for finiteness. Thirdly, there is evidence that Patty projects finite CPs, which on standard assumptions implies the presence of a T specified for finiteness.

The 'mapping' difficulty is a problem accessing morphological forms which have ‘layers’ of feature structure. Assuming a model of grammar where “an autonomous morphological component ‘reads’ the output of lexical and syntactic derivation, identifying those features which condition inflectional operations” (Lardiere 1998a: 20), Lardiere argues that the more inflectional features there are associated with a morphological form, the more likely a problem will arise for an L2 speaker. In the case of tense-marking, she assumes that the output of syntactic computations presents the morphological component with a terminal T node specified [+finite]. The morphological component must then determine whether it is [+past] or [−past], and if [+past] select suppletive forms in the case of irregular verbs, or invoke the regular rule for the affixation of *-ed* in the case of regular verbs. She proposes (speculatively) “that it is among the increasingly complex ‘outer’ layer mappings from morphology to PF that we are likely to find the greatest vulnerability to ‘fossilization’ and ‘critical period’ effects” (Lardiere 2000: 124). Additionally, if the phonological forms themselves involve complex phonology — for example, involving word final clusters like *-kt*, *-skt* and *-mpst* in the past tense forms of *walked, asked* and *glimpsed* — Lardiere argues that this may further affect successful mapping: “We can further imagine that an essentially morphophonological mapping procedure would be especially vulnerable to ‘derailment’ from a variety of post-syntactic or extra-syntactic factors, such as phonological transfer from the L1” (Lardiere 1998a: 21). Given that (Mandarin) Chinese is a language with basic (C)V(nasal) syllable structure and no syllable- or word-final consonant clusters (Hansen 2001), we can take Lardiere’s claim to be that while mapping of phonological forms onto terminal nodes which are the output of the syntax can cause problems generally for L2 speakers where layers of features are involved, in Patty’s case this is compounded by the fact that the L2 requires word-final consonant clusters where the L1 disallows them.

The combined results of the studies by Beck and Lardiere point to the following possible conclusion: a deficit has occurred in one of the mechanisms of the morphological component — the mechanism referred to as the ‘vocabulary’ in models of distributed morphology (Halle and Marantz 1993, Embick and Noyer 2001) — which inserts vocabulary items (phonological forms) into terminal nodes where the feature specification of the vocabulary item and the feature specification of the terminal node are non-distinct. The representation of morphological forms themselves is not affected (as Beck’s study suggests), and the feature specification of categories manipulated by the syntax is not affected, if Lardiere’s analysis is correct. Moreover, the difference in phonotactic constraints between the L1 and L2 has a persistent influence, such that the mapping problem is exacerbated. Thus this account of optionality in tense marking by L2 speakers is located at the interface between syntax and morphology.

In this chapter we test this claim by comparing the spontaneous oral production of advanced L2 speakers of English from three different L1 backgrounds: Chinese, Japanese and German. If there is a general mapping problem for L2 speakers at the interface between syntax and morphology involving feature matching at the point of vocabulary insertion, we would expect this to appear in all three groups. Since Japanese is similar to Chinese in its phonotactic structure (disallowing word-final consonant clusters) but German is like English (cf. word-final clusters such as *-ntst: getanzt* 'danced'), we would expect mapping problems to be more marked in the Chinese and Japanese informants.[2]

In fact we will argue that neither of the predictions is borne out by the data. The Chinese informants do mark simple past tense optionally in oral production, as previous studies had found, but the Japanese and German speakers are significantly less likely to do so. We will also show that our Chinese informants appear to know the morphological properties of English past tense verb inflection, as Beck's study suggests that they should, and that they do not generally appear to have problems producing word-final consonant clusters. This will lead us to argue for a different locus for their deficit: at the interface between the syntactic component and the lexicon. In particular, we will claim that the Chinese speakers have difficulty assigning the formal (i.e. syntactically-relevant) feature [past], which determines the morphophonological forms of verbs in English, to the feature inventory of the category T(ense) in the lexicon, because this feature is not selected in Chinese, and is subject to a critical period.

2. Assumptions about the organisation of the language faculty

We follow the spirit of recent work within the minimalist program (Chomsky 1998, 1999, 2001) and assume that the language faculty has a number of universally fixed and invariant computational procedures, and provides a universal inventory of phonological, semantic and syntactic (formal) features F from which lexical items can be assembled. One subset of the computational procedures is the syntax, which is capable of a small number of empirically and conceptually necessary operations: merge, agree and move. The syntax takes items presented to it from the lexicon and combines them into expressions. These expressions are interpreted by a semantic component (LF) (whose procedures are themselves universally invariant), and which makes the syntactic expressions 'legible' to the conceptual-intentional modules of mind. The syntactic expressions are also interpreted by morphological/phonological procedures which are universally uniform, and which make syntactic expressions 'legible' to the sensori-motor systems for the production and understanding of speech.

The universal inventory F of semantic, phonological and syntactic features is crucial in this model, because it is from this set that features are selected for the assembly of a lexicon whose items provide the input to the computational procedures. Individual languages select a subset of features from F and assemble them into lexical items. It is at this point — the selection of particular features for the assembly of lexical items — that languages vary.

We focus here on the selection of syntactic (formal) features. Syntactic features are those which initiate syntactic operations, for example, wh-movement, case agreement, N-to-D movement, and so on. Some selections of syntactic features appear to be obligatory. For example, finite T appears necessarily to select the syntactic feature required to activate structural nominative case. Nominals also appear to obligatorily select case features which render them active for the purpose of case agreement. Languages are uniform in selecting these features. Parametric differences between languages arise when they make different choices of optional syntactic features.

The distinction between obligatory and optional syntactic features is important for understanding tense and how it is realised morphologically in languages like English and Chinese. The view we adopt is that there is a syntactic (i.e. semantically uninterpretable) tense feature which, for the sake of exposition, we call [±past], which is available in the universal inventory F, but which is optional. English has selected it, but Chinese has not. In English, the presence of [±past] in finite T has a consequence for the morphology of the verb. [±past], being a syntactic (formal) feature, is not semantically interpretable at LF and so must be eliminated from a syntactic expression before such interpretation takes place. This elimination is effected through a checking (or matching) of the features of T with the morphological features of inflected verb forms like *was, had, walked, ran*. There are various complications that arise in this checking/matching operation depending on whether the verb is a light verb (*be, have, do*) or a thematic verb (*walk, run*). We will not expand on these here (see Lasnik 1999, Embick and Noyer 2001). But the basic claim is that finite English T has syntactic [±past] features which have morphological consequences for V. By contrast, we claim that Chinese does not have syntactic [±past] features on T, although it does have a syntactic [±finite] feature (Li 1990: 18).

As a consequence, bare Vs in Chinese can be interpreted either as past or non-past, depending on context:

(1) *Zhangsan kan dianying.*
Zhangsan see movie
'Zhangsan is seeing OR saw a movie.'

We take this up in more detail in Section 4.

3. The study

To test the claim that the source of optionality in tense marking by L2 speakers lies at the interface between the syntactic and morphological components of the language faculty, we selected advanced L2 speakers of English from different L1 backgrounds, devised a test aimed at measuring informants' knowledge of the morphological processes involved in simple past tense marking, and collected a sample of spontaneous production data including simple past tense verb forms from the same informants.

3.1 Informants

Advanced L2 speakers of English were selected for this study on the basis of their performance on an independent measure of general proficiency. This consisted of two components: (a) the written multiple-choice grammar test component of the Oxford Placement Test (Allan 1992) which has 100 items covering a range of the core morphosyntactic properties of English; (b) Nation's (1990) 'vocabulary levels test', which was designed as a language teacher's aid for giving help with vocabulary learning, but provides a rough notional measure of the size of a speaker's vocabulary up to the 10,000-word level. Using this combined grammar/vocabulary test, we selected informants whose L1 was Chinese, Japanese or German and whose mean scores broadly matched at the upper end (over 80% correct). This produced an experimental set of informants of two Chinese, five Japanese and five German speakers.[3] Details of the proficiency scores are given in Table 1.

3.2 Test of knowledge of morphology

If Beck's (1997) findings are generalisable, we expect to find that the informants in our study can productively manipulate morphological processes involved in past

Table 1. Proficiency test scores: experimental informants

L1	Mean proficiency score (%)	Range (%)
Chinese ($n=2$)	86.6	83.4–89.7
Japanese ($n=5$)	85.7	83.4–87.0
German ($n=5$)	90.7	85.8–96.0

tense marking in English. To test this we designed a task which required informants to inflect both real and invented (nonce) verb stems for simple past tense. Our reasoning was that if speakers know the verb morphology associated with past tense, it should make no difference whether real or nonce forms are involved.

3.2.1 *Design*

The task was adapted from one used by Prasada and Pinker (1993) with native speakers ('experiment 3' in their study). Prasada and Pinker were interested in the extent to which natives would produce regular and irregular past tense forms when presented with nonce verbs, and in particular whether novel irregulars displaying the 'prototypical' phonological shape of partially productive past tense irregulars (like *string, sling, fling, cling* → *strung, slung, flung, clung*) would elicit novel irregular past tense forms, such as *spling* → *splung*. To elicit such responses, informants were presented with nonce forms and an invented definition for each — six at the top of each page of a test questionnaire — followed by six sentences, each with a blank, where one of the nonce forms belonged (Prasada and Pinker 1993: 24).

In our test we were interested simply in whether informants would inflect verbs appropriately for past tense in clear past tense contexts, and whether their knowledge was generative in the sense of allowing them to inflect verbs they had never encountered before correctly. They were therefore presented with six verbs at the top of each page of a test questionnaire with definitions (as in the Prasada and Pinker study), but half were real and half invented (18 of them in fact taken from the set of prototypical regular and irregular nonce verbs used by Prasada and Pinker). Below each set of six verbs with their definitions were six sentences where informants had to decide which verb to insert, and what its form should be. These contexts required either a simple past or a present perfect form. Thus informants could not simply produce a past tense form by rote, but had to make a definite choice of tense appropriate to the context. A partial illustration of the form of the test is given in (2):

(2) SPLING: If you spling, you blow something out of your mouth (like smoke rings or air bubbles).
CUT: If you cut something, you make it shorter, or divide it, or break its surface.
a. The ground staff haven't marked out the tennis court or put up the net yet, but the head gardener claims that they __________ the grass in readiness.
b. As he rose slowly, the diver __________ bubbles in short bursts.

There were 120 contexts in all, 60 of which were expected to elicit simple past tense verb forms, and of these 30 involved nonce verb stems (15 prototypical regular types, like *blark*, and 15 prototypical irregular types, like *spling*). A control group of native speakers ($n=5$) also took the morphology test.

3.2.2 *Results*

Table 2 displays the frequencies of inflected and uninflected real and nonce verbs in simple past tense contexts.

A χ^2 test comparing the total frequency of inflected and uninflected forms produced by the non-native groups shows no significant difference between

Table 2. Frequencies of real and nonce verbs inflected for simple past

		L1							
		Chi ($n = 2$)		Jap ($n = 5$)		Ger ($n = 5$)		Eng ($n = 5$)	
Verb type	**Inflected**	**Score**	**(%)**	**Score**	**(%)**	**Score**	**(%)**	**Score**	**(%)**
Real reg	Yes	23	88.5	60	96.8	64	98.5	74	100
	No	3	11.5	2	3.2	1	1.5	0	0
Real irreg	Yes	29	100	64	97.0	65	100	73	100
	No	0	0	2	3.0	0	0	0	0
Nonce reg	Yes	25	92.6	58	96.7	64	100	72	97.3
	No	2	7.4	2	3.3	0	0	2	2.7
Nonce irreg	Yes	18	81.8	54	85.7	53	88.3	71	100
	No	4	18.2	9	14.3	7	11.7	0	0
Total	Yes	95	91.3	236	94.0	246	96.9	290	99.3
	No	9	8.7	15	6.0	8	3.1	2	0.7

NB: Cases where an informant selected a verb form other than simple past (e.g. past progressive or perfect) are excluded from the table. Hence scores do not necessarily add up to the expected maximum.

them ($\chi^2 = 4.94$, df=2, $p < .05$). A comparison between the non-natives as a group and the native controls shows that there is a significant difference ($\chi^2 = 12.64$ (with Yates' correction factor), df=1, $p < .05$). This difference appears to be located primarily in the extent to which the non-natives fail to inflect irregular nonce forms for simple past (where the native controls do inflect in 100% of cases). This is particularly interesting because there are no significant differences between non-natives and natives in inflecting regular nonce forms. The result seems to suggest that speakers know that certain irregular nonce forms are irregular, hence do not attach a regular inflection to them, but do not know what the inflected form should be, producing an uninflected form.

Broadly, and assuming that in some sense the non-native speakers are distinguishing nonce regulars from irregulars, frequency of past tense marking in the responses of these advanced non-natives is very similar to those of the natives. This is consistent with Beck's findings, and suggests that the morphological component is operating similarly in these speakers to the way it operates in natives.

3.3 Inflected simple past tense in spontaneous production

Spontaneous oral data were collected from two tasks: the retelling of a short extract from a Charlie Chaplin film (*Modern Times*), and the recounting of a happy or exciting experience each informant had had. The data were recorded and transcribed, and only verbs in unambiguously simple past tense contexts (i.e. those where a native could use no other form) were counted. For the purposes of this study, only thematic verbs were scored (e.g. *walked*, but not modals, copula/auxiliary *was* or auxiliary *had*). Verbs in contexts which were phonologically ambiguous were also discounted (i.e. regular past tense verbs followed by homophonic stops as in *walked to work*, or interdental fricatives, as in *chased them*). All unambiguous forms thus counted were rechecked against the original recordings to ensure accuracy. The frequencies of inflected and uninflected forms across the three groups of non-native speakers are presented in Table 3.

χ^2 tests show that there is a significant difference between groups both on frequency of inflection with regular verbs ($\chi^2 = 30.49$, df=2, $p < .01$) and with irregular verbs ($\chi^2 = 8.13$, df=2, $p < .05$). The difference appears to be located entirely in the Chinese speakers' performance. This is in contrast to the performance of these speakers on the morphology test, where there was no significant difference between the three non-native groups, either on real verbs or nonce forms.

Table 3. Frequencies of inflected/uninflected verbs in simple past tense contexts: spontaneous oral production

		L1					
		Chinese ($n = 2$)		**Japanese** ($n = 5$)		**German** ($n = 5$)	
Verb type		**Score**	**(%)**	**Score**	**(%)**	**Score**	**(%)**
Regular	Inflected	25	62.5	137	91.9	52	96.3
	Uninflect.	15	37.5	12	8.1	2	3.7
Total		40	100	149	100	54	100
Irregular	Inflected	64	84.2	252	93.3	79	95.2
	Ininflect.	12	15.8	18	6.7	4	4.8
Total		76	100	270	100	83	100

These results are problematic for the view that L2 speakers generally have difficulty mapping phonological forms ('vocabulary items') with 'layers' of morphological features onto terminal nodes generated by the syntax. If this were the case, we would expect all three groups to perform similarly on the spontaneous production task, but the Chinese speakers are performing significantly differently from the Japanese and German speakers.[4]

Is the difference the result of L1 phonology interfering with and depressing the performance of the Chinese informants? This is unlikely. Recall that the prediction was that if L1 phonology had such an effect, we would expect the Chinese and Japanese to experience similar problems because the relevant property (absence of word-final consonant clusters) is present in both L1s. However, to pursue this possibility further, we considered performance of the non-native speakers on consonant clusters elsewhere in their spontaneous production, specifically in monomorphemic words like *most, kind.* If word-final clusters are problematic in production, some evidence for this should surface in these forms. Table 4 compares informants' retention of *-t/-d* in inflecting regular past tense verbs with their retention of *-t/-d* in monomorphemes.

Frequencies are small, but suggestive. Although Chinese speakers do drop final *-t/-d* in monomorphemes, they do not do so as frequently as in the simple past. This result matches a similar finding in Bayley (1996), who in a group of 20 L1 Chinese speakers of intermediate and advanced proficiency in L2 English found a 65% retention of final *-t/-d* in monomorphemes versus a 44% retention in simple past tense verb forms. This is in marked contrast with the pattern

Table 4. Absence of word-final *-t/-d* in monomorphemes and regular simple past tense forms compared

		L1					
		Chinese		Japanese		German	
Word type	***-t/-d***	**Score**	(%)	**Score**	(%)	**Score**	(%)
Monomorphemes	Present	9	82	27	96	48	100
	Absent	2	18	1	4	0	0
Simple past (Regular)	Present	25	63	137	92	52	96
	Absent	15	37	12	8	2	4

of *-t/-d* deletion found in many studies of native speakers (Labov 1989), as already observed. Natives are more likely to drop *-t/-d* when a morphological boundary is not involved than when it is.

Another possibility is that spontaneous oral production introduces performance pressures which make it difficult for L2 speakers to access inflected past tense verb forms in real-time (Prévost and White 2000: 129). If this were the case, performance on the morphology test would be a better reflection of the informants' competence, because it lessens such pressures, while performance in spontaneous oral use of English underrepresents informants' competence. This also looks implausible in the context of the three-group comparison. If performance pressures were involved, we would expect them to surface in all three groups, not just in the Chinese speakers. However, to pursue the possibility further, we looked at performance of the informants in using regular inflected participles in cases like *were scared of*, *be sliced*, *is released*, which are identical to the simple past tense forms. Table 5 brings together the frequencies of inflected/uninflected regular participles, monomorphemes and regular simple past tense forms.

Again, although small frequencies are involved, if performance pressures were responsible for producing optionality in simple past tense marking, we might expect to see some evidence in the production of participles, where layers of morphological features also seem to be involved: if a verb form is [−finite] then it is either [+durative] (*walking*) or [−durative]. If it is [−durative] it is either [+past] (*(have) walked*) or [−past] (*(to) walk*).

In summary, the results from spontaneous oral production are the following: although the non-native groups were matched for general high proficiency in L2 English, the Chinese informants were significantly less likely to inflect

Table 5. Absence of word-final *-t/-d* in regular participles, monomorphemes and regular simple past tense forms compared

		L1					
		Chinese		Japanese		German	
Word type	*-t/-d*	Score	(%)	Score	(%)	Score	(%)
Participles	Present	10	100	23	100	55	100
	Absent	0	0	0	0	0	0
Monomorphemes	Present	9	82	27	96	48	100
	Absent	2	18	1	4	0	0
Simple past (Regular)	Present	25	63	137	92	52	96
	Absent	15	37	12	8	2	4

both regular and irregular thematic verbs for past tense than the Japanese or German speakers. The Chinese speakers retained word-final *-t/-d* with monomorphemes more often than with regular past tense verb forms (suggesting the problem is not caused by word-final consonant clusters). The Chinese speakers were perfect in inflecting past participles, in contrast to simple past tense verb forms, suggesting that 'performance pressures' are unlikely as a source of omission of inflections with past tense verbs.

4. Discussion

We are interested in locating the source of optionality in the marking of thematic verbs for simple past tense in oral production by certain groups of high proficiency L2 speakers of English, for example speakers of L1 Chinese. To address this problem comparative data were collected from L1 speakers of Chinese, Japanese and German matched for general proficiency as 'advanced' speakers of English. An important caveat in discussing the results is that the number of informants was small, therefore the generalisability of any conclusions drawn must be treated with caution. The data were elicited from a test of knowledge of past tense morphology, and tasks allowing free oral production (an anecdote and the retelling of a film). Results of the morphology test suggest that all three groups, including the Chinese speakers, know the morphological properties of past tense marking in the context of real English verbs, and are productively aware of the regular versus irregular distinction even with verbs they have never encountered before (nonce forms). This is consistent with earlier findings of Beck (1997), who showed that the reaction times of non-native speakers on a verb inflection task were parallel to those of native speakers on the same task. This led her to conclude that the source of optionality of past tense marking is not located in the operation of the morphological component, a claim with which we concur.

The comparative results from the free oral production of our three experimental groups show that the Chinese speakers are significantly different from the other two groups, producing uninflected regular verbs in unambiguously past tense contexts in over one third of cases. This was *prima facie* evidence against the idea that L2 speakers in general have difficulty mapping phonological forms ('vocabulary items') onto syntactic terminal nodes where that mapping implicates layers of morphological structure. If this were a general problem for L2 speakers, and on the assumption that morphological properties do not transfer from the L1 to the L2, we expected all three groups to show evidence of it. By looking at the Chinese speakers' performance on word-final consonant clusters in monomorphemic words, and their performance in inflecting past participles, we also established that optionality in producing final *-t/-d* was lesser in these contexts. This suggested that the source of the problem with simple past tense was unlikely to be phonological in nature or the result of performance pressure because similar optionality would be expected across these contexts too.

Having eliminated the morphological component, and the mapping of morphophonological forms onto syntactic representations as likely sources of observed optionality in past tense marking in the informants studied, an alternative explanation is needed. In this final section we explore the consequences of assuming that optionality results from a failure of L2 learners to include a syntactic feature for tense, that we are calling [±past], among the features which make up the lexical item T. Where this is the case, T enters syntactic derivations without the feature which, for native speakers of English, eventually forces the insertion of vocabulary items inflected for past tense into terminal nodes. Given such a hypothesis, we need to account for why optionality is characteristic of the Chinese speakers in our sample, but not the Japanese or German speakers; why the Chinese speakers nevertheless have greater than chance success in marking verbs for past tense in past tense contexts; and why the Chinese speakers might be more successful in marking irregular past forms and regular participles than in marking regular past forms. Central to this account is an understanding of the relation between how tense is interpreted by the semantic component, and the presence of a syntactic (formal) tense feature in T. We therefore examine this relationship more closely.

Languages appear to differ in whether they 'grammaticalise' tense through special morphophonological forms or not. Chinese is standardly assumed to lack such morphophonological forms, although it does have verbal aspect markers like *-le*, *-guo* and *-zhe* (Li and Thompson 1981, Li 1990, Packard 2000). For example, as already noted in Section 2, the verb *kan* 'see' can be freely interpreted as present or past depending on context:

(3) a. *Zhangsan kan dianying.*
Zhangsan see movie
'Zhangsan is seeing a movie.'
b. *Zhangsan zuotian kan dianying.*
Zhangsan yesterday see movie
'Zhangsan saw a movie yesterday.'

By contrast, Japanese and German do appear to grammaticalise tense, like English. Japanese has the forms *-ta* (past) and *-ru* (non-past) which appear to be tense auxiliaries (Okuwaki 2000). Thematic verbs in German inflect for past versus non-past tense like English, with regular and irregular variants (e.g. regular: *kaufen* 'buy', *ich kaufte* 'I bought'; irregular: *singen* 'sing', *ich sang* 'I sang').[5]

Why might some languages have grammatical exponents of a 'past/non-past' distinction while others, like Chinese, simply lack such exponents? An interesting perspective on this question can be found in the work of Chierchia (1998). Chierchia suggests that the presence of a syntactic property in a grammar which is associated with a particular semantic operation has the effect of inhibiting the free application of that operation. In his own study he suggests that the presence of articles in a language blocks the free application of the semantic operations which give nominals generic, definite or indefinite meanings. So in English, count Ns can only be interpreted as definite when *the* is present, but in Russian, which lacks articles, bare Ns can be freely interpreted as generic, definite or indefinite 'depending, presumably, on the context' (Chierchia 1998: 361). This idea has been extended by Takeda (1999: 103) into a Generalised Blocking Principle (GBP): if a language has a certain functional category in its lexicon, the free application of the semantic operation that has the same function as that syntactic category is blocked in that language.

Taking 'functional category' here to mean 'feature of a functional category', what the GBP proposes in terms of tense is that the semantic operation which interprets a T-V configuration as past or non-past can apply freely except where a language assigns a syntactic [±past] feature to T. This then requires overt morphological expression and blocks the free operation of tense interpretation. So while finite bare Vs in Chinese can potentially be interpreted as past, present or future, depending on the context, finite bare Vs in English can only be interpreted as non-past, because past tense interpretation is associated with the specific features which give rise to forms like *walked, ran*.

Consider now how this idea might be implemented in a grammar which incorporates some version of distributed morphology (Halle and Marantz 1993, Embick and Noyer 2001), the kind of model assumed in the work of Lardiere (2000) and Prévost and White (2000). The syntactic component generates expressions from bundles of syntactic and semantic features using the operations merge, move and agree (Chomsky 1998, 1999, 2001). Expressions consist of strings of terminal nodes which are presented to the morphological component for the insertion of vocabulary items (phonological forms). Insertion is an automatic process which takes place where the features of a vocabulary item are non-distinct from the features of a terminal node. For example, the terminal string T-V will present the morphological component with features like [+finite, −past] or [+finite, +past]. Vocabulary items with the relevant features will then compete for insertion into this position. Two important properties of insertion are firstly that a vocabulary item does not necessarily have to have all of the features of the terminal node to be inserted; it is sufficient that the features of the item are non-distinct from those of the terminal node (i.e. they may be a subset of those features). Secondly, that where vocabulary items are in competition for insertion into a terminal node, the most highly specified item compatible with the features of the terminal node is the one which wins the competition for insertion (Lumsden 1992: 480). For example, suppose that variants of *walk* have the following feature specifications:

(4) walks [V, +finite, 3p, +sing]
walked [V, +finite, +past]
walk [V]

Given such specifications, only *walked* can be inserted into a terminal node with the features [+finite, +past], even though *walk* is unspecified for those features and is in principle available for insertion. The reason is that although both forms have feature specifications which are non-distinct from the terminal node, *walked* is the more highly specified item.

With such an account of the interaction between syntax and morphology, it is easy to see how the proposal that adult L2 speakers have difficulty with the mapping of morphophonological forms onto terminal nodes might be made explicit: the procedure for inserting more specified vocabulary items over less specified items is not operating categorically in L2 speakers; less specified forms are being inserted where they should not be.

The account we wish to explore here, however, is one where the syntactic feature [±past] is absent from T in the terminal string which is the output of the syntactic computations. The claim we will make is the following: where parametrised syntactic features are not present in a speaker's L1, they will not be accessible in later L2 acquisition.

This is equivalent to saying that syntactic features which invoke the Generalised Blocking Principle that have not been activated in early life do not operate in adult SLA. This immediately distinguishes Japanese and German speakers on the one hand, from Chinese speakers on the other. Japanese and German both have morphosyntactic exponents of 'past tense', hence the underlying syntactic features which invoke the GBP. In principle this should allow them to determine that past tense verb forms in English are syntactically motivated. By contrast, for L1 speakers of Chinese, T does not have the syntactic feature [±past]. Our proposal entails that when Chinese speakers learn English they are unable to establish that English T is specified for [±past]. This would mean that in their English grammars the terminal string T-V will include [±finite] but not [±past]. The distinction between vocabulary items like *walks, walked, walk* would then have no syntactic motivation for them.[6] This is one kind of answer to the first question: why is optionality in past tense marking characteristic of the Chinese speakers in our sample, but not the Japanese and German speakers? It is also clear, however, that the Chinese informants studied here have acquired vocabulary items with past tense forms, and use them (at least superficially) in a highly target-like way in the morphology test, and at above chance level in spontaneous production. What might explain this if our claim is correct?

One possibility is that Chinese speakers analyse participles and irregular past tense verb forms differently from regular past tense forms: they have a different morphological status in their grammars. Past participles are arguably different from simple past tense forms even in native grammars, in that they are aspectual in nature (realising 'perfectivity'), and result from a verb-internal word formation process, which does not involve the T-V configuration. Since our assumption is that L2 speakers do not have difficulty with morphological operations *per se*, it would not be surprising if they did not have difficulty with participle forms. In other words, we expect Chinese speakers to have difficulty with simple past tense forms because they involve a syntactic feature missing from T, but not participle forms because they do not involve T. In the case of irregular past tense forms, one possibility is that the Chinese speakers have acquired them as items independent from the equivalent bare V forms; i.e. *ran* is not the past tense exponent of *run* as it is in native grammars, but an independently acquired word form, in the same way that, say, *amble* and *saunter* are independent word forms for native speakers. While this is speculative, there is some limited evidence consistent with it in the transcripts of the spontaneous production data. There are some cases of 'doubly inflected' verb forms such as in (5a). The same speaker who produced (5a) also used *ran* in clearly non-past environments ((5b) and (5c)):

(5) a. The girl *ranned* not far away.
b. You should *ran* away together.
c. She could not *ran* any more.

If *ran* were an independent word form it could be inflected for past tense and be used in non-past contexts. Observe also that (5b) and (5c) involve the use of *ran* as a non-finite verb. Given the model of vocabulary insertion assumed, *ran* could not be an inflected variant of *run* specified for the feature [+past]. If it were, its feature specification would clash with the feature specification of the non-finite terminal nodes into which it is inserted in (5b) and (5c). Insertion should simply be impossible.

If past participles and irregular forms like *ran* have a different morphological status in the L2 English grammars of Chinese speakers from regular past tense verb forms like *walked*, this would be one kind of answer to the question: why are Chinese speakers more successful in marking irregular past tense verbs and regular participles than regular past tense verbs? They are not treating them as past tense verb forms at all; rather they have independent lexical item status for Chinese speakers. However, given this account, we have little idea currently of what these forms might 'mean' for Chinese speakers.

This leaves the need to give an account of optionality in the marking of regular thematic verbs for past tense. We have no real answer to give in this case. Firstly, we have to assume that there is some reason why Chinese speakers do not treat regular verbs as independent vocabulary items as in the case of irregulars. One possibility is that it is the very regularity of the inflection and the frequency of such forms in the input which forces a morphological analysis of them as rule-based variants of the bare V. The problem then is to explain why some forms, but not others, are inflected in spontaneous production. To give a flavour of the problem, consider the following short extract from the transcript of one of the Chinese informants in our sample:

> "When I *saw* the film 'Lonely and Hungry' and it *reminded* me of the old time when life *was* very hard. Some people they *were* very hungry and they *have* no work to do. They really *don't* want to steal. But they *had* no other choice and when they *become* so hungry and they really *want* to just get the food and just *want* to eat the food so they *stole* or they just *do* something bad. But it'*s* understandable. It *was* not their mistake … And I *watch* it maybe 20 years ago. But it … I *didn't* remember clearly about what it *talk* about. I just *laugh* a lot. But now when we *see* the film, I think. It *gave* me much imagination."

What might be prompting this speaker's use of the bare verb forms *want, watch, talk* and *laugh* in a context where past tense reference is often clearly intended? We have little idea yet of what the answer might be, but some possibilities appear unlikely. One explanation for selective verb inflection that has been advanced in studies of the early stages of L2 acquisition is the 'Aspect Hypothesis'. This maintains that English past markers in early grammars are associated preferentially with verbs/predicates which have 'telicity' as part of their meaning ('achievements' and 'accomplishments' in the terminology of Vendler (1967)). See Bardovi-Harlig (1999) for a review of work on this topic. Does the same pattern obtain in the high proficiency Chinese speakers investigated here? That is, are they treating the regular past tense form as a marker of inherent verbal/predicate aspect? A breakdown of inflected regular forms by verb type is given in Table 6.

Table 6. Inflected past tense regular verbs by aspectual type: Chinese speakers

	Statives	Activities	Accomplishments	Achievements
Inflected tokens	0/4	10/14	1/2	14/20
%	0%	71%	50%	70%

The results are not conclusive. While statives are not inflected at all, which would be consistent with the Aspect Hypothesis, activities (which are atelic) are inflected to the same degree as achievements (which are telic). Interestingly, Lardiere (2002) has also analysed Patty's past tense verb forms in terms of whether there is a correlation with the telicity of the verb/predicate, and found that 40% of all telic verbs (112/277, covering both regulars and irregulars) were marked for past tense, while 35% of all atelic verbs (130/371) were past-marked.

This is a non-significant difference (using χ^2). Thus, although the Aspect Hypothesis might be an explanation for the distribution of English past tense verbs in low proficiency L2 speakers, it looks unlikely as an explanation for optionality in the marking of past tense on regular verbs by high proficiency L2 speakers.

Another unlikely possibility considered (and rejected) by Lardiere in the case of Patty is the 'Discourse Hypothesis' (Bardovi-Harlig 1995). This holds that L2 speakers of English may initially use past forms of verbs to mark foreground events in narratives, but not mark background events. 'Foreground' events are defined as clauses that move time forward in a narrative and which, if interchanged, would change the sequence of events in the narrative; 'background' events are often 'out of sequence', provide additional information about the foreground events, set the scene, evaluate or explain (Bardovi-Harlig 1995: 265–267). Lardiere calculates that 32% of Patty's past-marked verb forms (13/41) describe foreground events, while 30% (11/38) describe background events.[7] Again, while the Discourse Hypothesis might explain the distribution of forms in low proficiency L2 speakers, it looks unlikely to be the source of optionality in high proficiency L2 speakers.

The only hypothesis we are able to advance at present (but for which there is scant evidence in our current sample) is the following: linguistic theory appears to need to allow operations which apply to strings post-syntactically, that is in the morphological component or following vocabulary insertion. For example, Chomsky (1999) proposes that English has an output condition which bars surface adjacency between V and direct objects where the V is an unaccusative or a passive. For example, although (6a) is the expected output expression generated by the syntax, it is less natural than (6b) or (6c). Chomsky claims that a post-syntactic output constraint forces the object to move either leftwards or rightwards:

(6) a. There was placed a large bowl on the table.
b. There was a large bowl placed ___ on the table.
c. There was placed ___ on the table a large bowl.

The surface ordering of clitic clusters in French also appears to be the effect of a post-syntactic output condition (Perlmutter 1971). If two third person clitics co-occur, an accusative form precedes a dative, but if a first or second person and a third person clitic co-occur, a dative form precedes an accusative (as in (7)):

(7) a. *Elle le lui donne.*
she it-ACC him-DAT gives
'She is giving it to him.'

b. *Elle me le donne.*
she me-DAT it-ACC gives
'She is giving it to me.'

This suggests that the language faculty allows some monitoring of surface strings. Perhaps in the normal case for Chinese speakers the morphology inserts bare verb vocabulary items into T-V strings. But because output checking is a possibility available in the grammar, Chinese speakers monitor the ambient discourse for 'pastness' and insert V-ed forms when they are able to detect it. So, as in French where ordering of object clitics is determined by the person of the forms in question (first and second person must precede third person), for the Chinese speakers selection of a V-ed form is determined by 'pastness'. The difference between the two cases is that whereas [person] is a feature present in the specification of the French pronoun vocabulary items, 'pastness' has to be determined on the basis of context.

Because context is involved, the monitoring process is unstable and gives rise to the kind of apparently random use of bare and inflected verb forms illustrated in the sample of informant speech given above. This is consistent with two observations about the marking of regular verbs for past tense by Chinese speakers. First, individual speakers of high proficiency can differ markedly in the extent to which they are successful in inflecting verbs. Our informants are apparently more successful in spontaneous oral production than Patty, for example. Secondly, different modalities allow a greater degree of success in past tense marking by the same individual. So our informants were more successful on the morphology test than in oral production, and Lardiere (2002) reports that Patty's written output (in the form of e-mail messages) shows higher proportions of past tense marking than her spoken output.

5. Conclusion: The interface between syntax and the lexicon in adult SLA

In a comparison of the performance of three groups of highly proficient L2 speakers of English (with Chinese, Japanese and German as L1s), it was found that the marking of past tense was optional in the spoken English of the Chinese speakers, but not in the case of the Japanese and German speakers. With the caveat that caution is required in generalising from small numbers of informants, we argued that the Chinese speakers' knowledge of English morphological processes is intact (as Beck (1997) had already argued), and that phonological properties (i.e. *-t/-d* deletion) did not appear to be a factor for our speakers. Furthermore, we found that past participle and irregular past tense forms were inflected more consistently than regulars.

Although previous studies have argued that L2 speakers can fully acquire the syntactic features of lexical items like T (Lardiere 1998a, 1998b, 2000, Prévost and White 2000), we explored the alternative possibility that Chinese speakers cannot establish [±past] on T in English precisely because this feature is absent in their L1. By contrast [±past] is present on T both in Japanese and German. If Chinese speakers do not have access to such a feature in the construction of L2 knowledge, but Japanese and German speakers do, this would explain the observed difference between them in inflecting thematic verbs for past tense in spontaneous production.

We linked this 'deficit' in the Chinese speakers' grammars to the idea that optional syntactic features, when selected, 'block' the free application of semantic operations (Chierchia 1998, Takeda 1999). Our account assumes that these optional properties are subject to a critical period (an idea which originates in the work of Tsimpli and Roussou (1991) and Smith and Tsimpli (1995)). We then explored, in highly speculative mode, reasons why Chinese speakers might be successful at all in marking thematic verbs for past tense.

This line of enquiry raises an interesting possibility for the interface between the syntactic component and the lexicon in adult SLA. Within the spirit of 'minimalist' enquiry, it might be proposed that all of the resources and computational procedures of the language faculty — LF, syntax, morphology — are intact and operative in adult SLA. However, optional (parametrised) syntactic features which play a major role in tying morphosyntactic structure to semantic operations in the L1 acquisition of specific languages are unavailable if not activated in early life. In parsing L2 data, older L2 speakers can use all of the resources except for these features. Phonological forms ('vocabulary items') which are selected by such features in native grammars are not selected by those features in the grammars of L2 learners beyond the critical period. Highly proficient speakers must nevertheless find some motivation for them. We rejected some of the conceivable ways in which Chinese speakers might motivate past tense verb forms in English (the 'Aspect Hypothesis' and the 'Discourse Hypothesis'), and we suggested that they might be operating in terms of an 'output condition' which monitors the T-V string in relation to the 'pastness' of the discourse.

Notes

* Parts of this work have been presented to audiences at McGill (2000) and GASLA (2000). We are grateful for the comments of those present in both cases. We have also benefited greatly from the comments of Donna Lardiere on an earlier draft discussion of the ideas outlined here. She will not agree with our conclusions, but her views have helped us sharpen our thinking on a number of issues.

1. The phenomenon of *-t/-d* deletion from word-final consonant clusters attested widely in informal varieties of native English (Labov 1989), is strongly disfavoured in the context of the regular simple past tense (Bayley 1996: 109). We return to this below, since it appears that the pattern is the reverse in non-native speakers: *-t/-d* is more likely to be missing from the regular past tense forms of verbs than in other contexts.

2. The assumption underlying the predictions made in this paragraph is that the morphological features of vocabulary items do not transfer from the L1 to the L2. Donna Lardiere (p.c. and 2000: 116–117) points out that the assumption is not self-evident. If the "L1 exhibits the same (or higher) degree of [morphological] complexity [L2 learners] may know what to look for". Whether L2 speakers do or do not transfer morphological properties from their L1 is obviously an empirical question. If it turns out that they do, the observations reported in this article will need to be reinterpreted. Given that evidence bearing on the empirical question is currently lacking (although see Jarvis and Odlin (2000) for some relevant discussion), for the purposes of this article it will be assumed that transfer of verbal morphological properties from L1 to L2 is not a factor.

3. The small size of the Chinese group is a reflection of the difficulty we had in locating L1 Chinese informants who can achieve a score of 80% or above on the proficiency test (rather than a difficulty in finding L1 Chinese speakers of L2 English per se). Since, however, for the purposes of the present investigation, the frequency of past tense marking in non-native-speaker English is the focus of interest, the variation in group size is relatively unimportant. In further work it would be desirable to have larger samples of advanced English speakers of the L1s in question.

4. Unless, of course, L2 speakers whose L1s are morphologically complex 'know what to look for'. See footnote 2.

5. However, it should be noted that the present perfect — *ich habe gekauft* — predominates in everyday spoken German in what would be simple past contexts in English (Comrie 1976).

6. A question that might be raised here is whether this shouldn't predict random usage by the Chinese speakers of all three forms (Donna Lardiere p.c.). This would be the case only if L2 speakers assumed that morphological variants of a lexical item can occur in free variation. However, we argue subsequently that Chinese speakers attempt to establish representations which distinguish morphologically distinct forms. The problem for them is that they have to do so without the benefit of the syntactic feature [±past].

7. In practice the criteria seem to allow for considerable variation between raters in deciding which events are in the foreground and which in the background. We had difficulty applying them to our own sample, and do not report the results here.

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